Total No. of Pages: 2 *Roll No.....*

VITH SEMESTER B.Tech.(Computer Engineering)

MID SEMESTER EXAMINATION (March, 2020)

CO304 ARTIFICIAL INTELLIGENCE

Paper Code

Time: 1:30 Hours Max. Marks: 25

Note:	Answer all questions.
	Assume suitable missing data, if any.

Q1. Answer following with suitable reasoning:

(i) Is Breadth first search is intelligent search procedure?

(ii) Are Database and knowledgebase same?

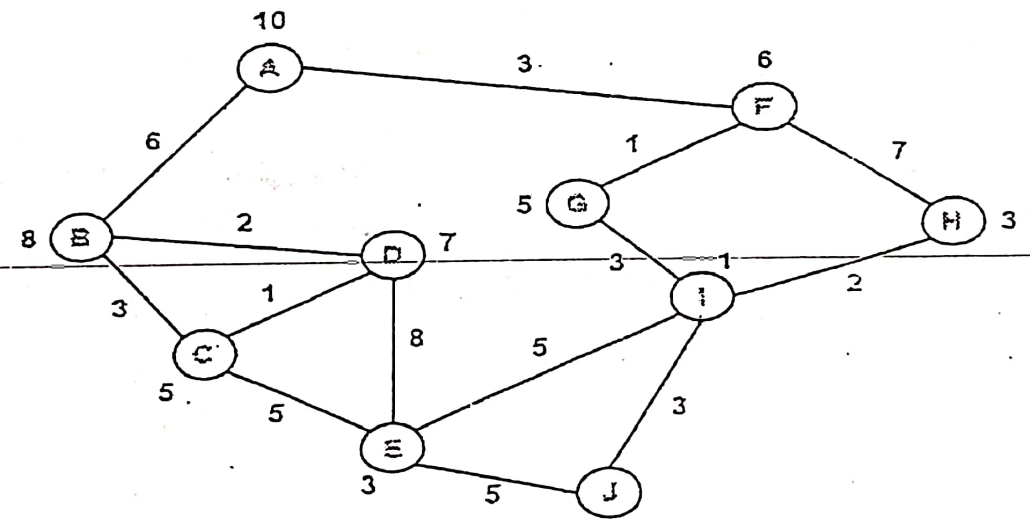
(iii) Is Best First Search better than Hill Climbing?

(iv) Does A* always find optimal solution?

[4x2]

Q2. Analyze the project characteristics of 8 puzzle problem. [2]

Q3. Consider the OR graph shown below, where number written on edges represent the distance between the nodes. The number written on nodes represent the heuristic values.



Find the most cost-effective path to reach from start A to final state J using: (i) A* (ii)Best first search algorithm. Explain steps used. [4]

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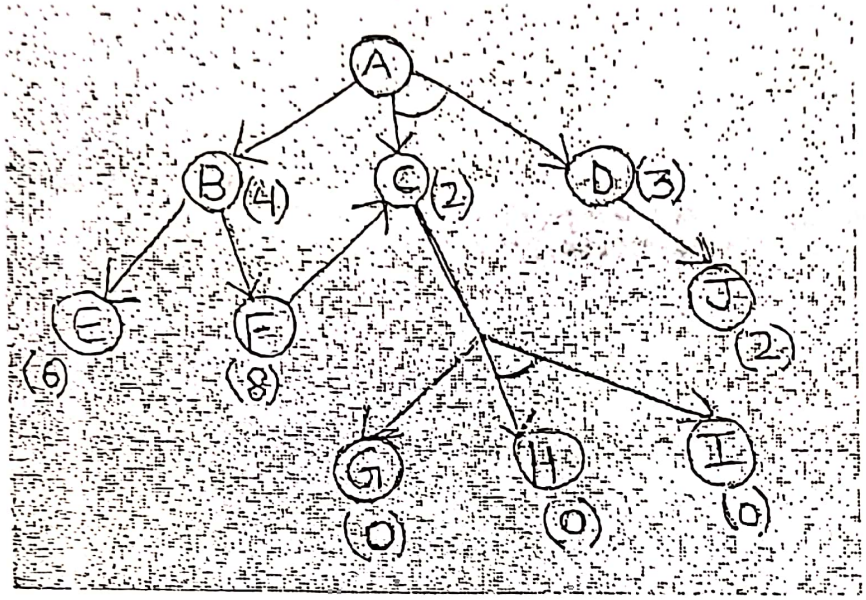
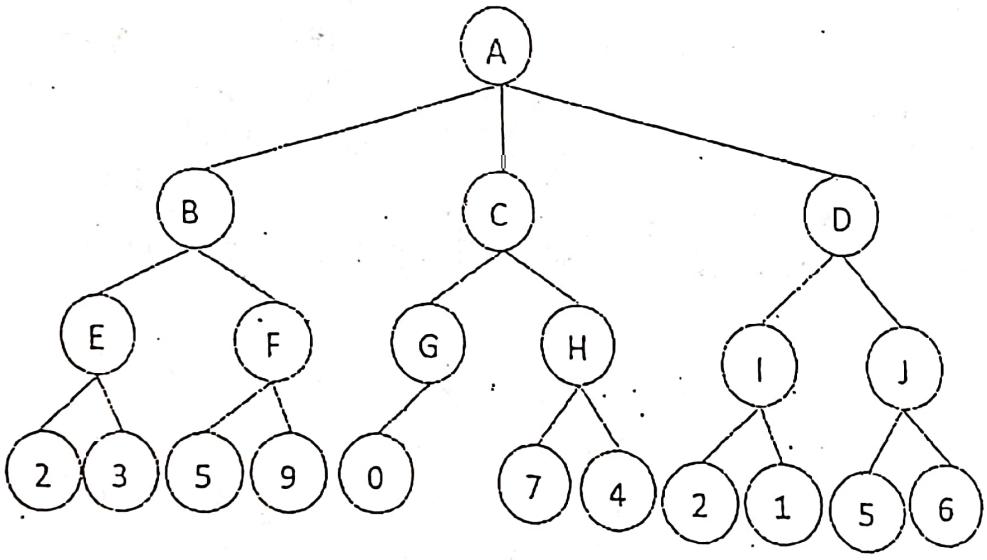
Q4. Consider the AND OR graph shown below. Values in bracket represent heuristic values Work out the steps of AO* procedure on this graph and find out the path from Starting node A to goal state, where heuristic value is zero [3]



Q5. For the graph below, apply min-max procedure to decide move for Player MAX. What is the max value at the root node by applying mini-max search? Also show alpha beta pruning [4]



Q6. What are the different Knowledge representation methods available? What are the essential characteristics of any knowledge representation method ? [4]

(ii)